

Bahrouni Ayoub

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SUMMARY

Software and IoT Developer with a diploma in **Computer Systems Engineering – IoT and Embedded Systems** and a certification in **Full-Stack Development**. Skilled in building end-to-end solutions, from **embedded systems** using ESP32 and MQTT to **web and mobile applications** using React, Node.js, MongoDB, and Flutter. Experienced in integrating hardware, cloud, and software layers to create scalable, user-centric applications. Passionate about innovation, problem-solving, and delivering impactful technology solutions.

EXPERIENCE

IoT & Software Development Intern

Tunisie Télécom

March 2025 – June 2025, Ariana, Tunisia

- **Engineered** a smart irrigation system integrating IoT sensors, ESP32 microcontrollers, and AI models for predictive water management.
- **Developed** a cross-platform **Flutter mobile app** and a **Node.js + MongoDB backend** for real-time data visualization and analytics.
- **Integrated** MQTT protocols to ensure reliable, low-latency device-to-cloud communication and optimized system responsiveness.
- **Enhanced** operational efficiency by leveraging predictive analytics to reduce water waste and improve irrigation scheduling.
- **Documented** system architecture and deployment procedures, ensuring scalability and future maintainability.
- **Key Skills Demonstrated**
 - IoT Development (**ESP32, MQTT, sensors**)
 - Mobile Development (**Flutter**)
 - Backend Development (**Node.js, MongoDB**)
 - Frontend Development (**React.js**)
 - Cloud Integration and Optimization (**scalable storage, real-time analytics**)

PROJECT

PFE – AI-Driven Smart Irrigation

Tunisie Télécom • github.com/BahrouniAyoub/PlantCare_Flutter.git • March 2025 – June 2025

- Designed and implemented a **smart irrigation system** integrating IoT sensors, ESP32, and MQTT for real-time monitoring.
- Built a **Flutter-based mobile app** and **Node.js + MongoDB backend** to visualize data and optimize water usage.
- Integrated basic AI models to predict irrigation needs, improving efficiency.
- **Skills Used:** ESP32, MQTT, Flutter, Node.js, MongoDB, React.js, AI basics

EDUCATION

Bachelor's Degree in Computer Systems Engineering – Specializing in IoT and Embedded Systems

Minor in Full-Stack development • Faculty of Sciences of Tunis • Rue de Tolède, Tunis 2092 20 • 2025

CERTIFICATIONS

Full-Stack Development

3w Academy • 2023

- “The 3W Academy certificate gave me **hands-on experience in building full-stack applications** with modern frameworks like React, Node.js, and MongoDB. These are skills that directly support developing scalable and user-focused software, which complements my IoT and embedded systems background.”

COURSEWORK

Networking and Communication Systems

Faculty of Sciences of Tunis • 2025

- Understanding TCP/IP stack and wireless communication. Designing and testing network topologies. Basic network security principles.
- Configured IoT devices for stable network communication during testing phases.
- Simulated a **Smart House network** using Cisco Packet Tracer to model device interactions.

Mobile App Development (Flutter, Android Studio)

Faculty of Sciences of Tunis • 2025

- Cross-platform mobile UI design and API integration. State management and user authentication. Testing and deploying mobile applications.
- Built the **Flutter-based mobile app** for your PFE, integrating real-time IoT data visualization.
- Developed an **Android health app** with secure authentication and a responsive interface.

Embedded Systems Programming (C, C++)

Faculty of Sciences of Tunis • 2024

- Low-level programming for microcontrollers. Memory management and hardware interfacing. Debugging and optimizing embedded code.
- Programmed **STM32 and ESP32 microcontrollers** to collect and process sensor data.
- Developed **real-time control logic** for your Smart Garage Door project.

IoT Architecture and Protocols (MQTT, HTTP, TCP/IP)

Faculty of Sciences of Tunis • 2024

- Designing communication layers for IoT systems. Using lightweight protocols for efficient data transfer. Securing and optimizing device-to-cloud communication
- Integrated **MQTT protocol** in your Smart Irrigation PFE to ensure reliable real-time data exchange.
- Used **HTTP APIs** for data transmission between devices and cloud services in other prototypes.

Microcontroller Programming (ESP32, STM32, Arduino)

Faculty of Sciences of Tunis • 2024

- Configuring microcontrollers and peripherals. Sensor calibration and integration. Power management and hardware troubleshooting.
- Programmed **ESP32 boards** to monitor soil humidity and temperature in your PFE.
- Built sensor-driven logic for your **Smart Garage Door** and **home automation simulations**.

Database Design and Management (SQL, NoSQL)

2023 • Database schema design and normalization. CRUD operations and query optimization. Using NoSQL for unstructured IoT data.

- Implemented **MongoDB backend** for your Smart Irrigation and health app projects.
- Designed relational databases for a **Library Management System** in Java.

SKILLS

Software: TypeScript, Next.js, Express.js, React, PHP, SQL, NoSQL, MongoDB, SQLite, MySQL, Data Security, C++, Java, JavaScript, C#, Python, Embedded C, Flutter, Node.js, STM32, ESP32, Raspberry Pi, HTML, CSS, REST APIs, Flutter, Firebase, Supabase, MQTT, WebSocket, HTTP/HTTPS, JWT, TCP/IP, MERN, PostgreSQL
